

1.0. INTRODUCTION

AIM: Getting started with MATLAB

1.1. Introduction

MATLAB is a powerful language for technical computing. The name MATLAB stands for MATrix LABoratory, because its basic data element is a matrix (array).

MATLAB can be used for math computations, modeling and simulations, data analysis and processing, visualization and graphics, and algorithm development. It integrates numerical analysis, matrix computation, signal processing and graphics in an easy-to-use environment, where problems and solutions are expressed just as they are written mathematically, without traditional programming.

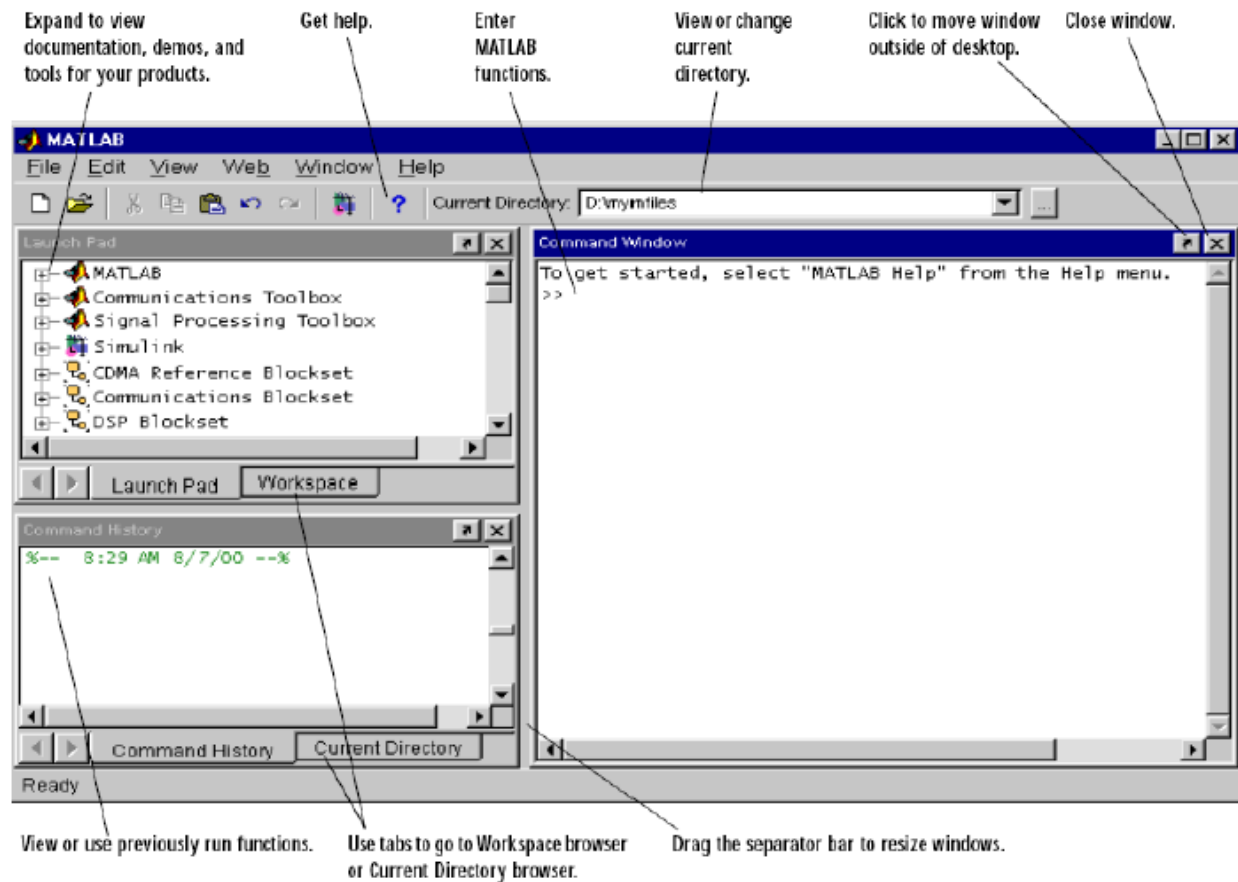
The standard MATLAB program has tools (functions) that can be used to solve common problems. It also has optional toolboxes that are collections of specialized programs designed to solve specific types of problems.

Examples include toolboxes for signal processing, symbolic calculations, and control systems.

Steps

- Start MATLAB by double-clicking on the MATLAB shortcut *icon* on your Windows desktop or from all programs on start button.
- When you start MATLAB, a special window called the MATLAB desktop appears. The desktop is a window that contains *other* windows. The major tools within or accessible from the desktop are:
 - The Command Window
 - The Command History
 - The Workspace
 - The Current Directory
 - The Help Browser

Explain the main windows in matlab desktop?



This illustration also shows the default configuration of the MATLAB desktop. You can customize the arrangement of tools and documents to suit your needs.

1.2. Guidelines to Working with MATLAB

- Files are saved as .m files.
- To add comment the following symbol is used "%".
- Help is provided by typing "help" or if you know the topic then "help function_name" or "doc function_name".
- If you don't know the exact name of the topic or command you are looking for, type "look for keyword" (e.g., "look for regression")
- Three dots "..." are used to continue a statement to next line (row).

- If after a statement “ ; ” is entered then MATLAB will not display the result of the statement entered otherwise result would be displayed.
- Use the up-arrow to recall commands without retyping them (and down arrow to go forward in commands).
- MATLAB is **case sensitive** so “ab” is not same as “AB”
- MATLAB utilizes the following arithmetic operators;
 - + Addition
 - - Subtraction
 - * Multiplication
 - / Division
 - ^ Power Operator
 - ' transpose
- Some built in functions in MATLAB
 - **abs** magnitude of a number (absolute value for real numbers)
 - **angle** angle of a complex number, in radians
 - **cos** cosine function, assumes argument is in radians
 - **sin** sine function, assumes argument is in radians
 - **exp** exponential function

1.3. Basic Operations

Simple operation $1 + 2 \times 3$

```
>> 1 + 2 * 3
```

```
ans =
```

```
7
```

Using Semicolon

```
>> x = 1 + 2 * 3;
```

```
>> x
```

```
x = 7
```

Entering multiple statements per line

```
>> a = 7; b = cos (a),  
b = 0.6570
```

Defining a column vector

```
v = [1;2;3]  
v =  
1  
2  
3
```

Defining a row vector

```
w = [1 0 1]  
w =  
1 0 1
```

Transpose a vector

```
W = w'  
W =  
1  
0  
1
```

Defining a range for a vector

```
>> X = 1: .5 : 5  
  
X =  
  
1.0000 1.5000 2.0000 2.5000 3.0000 3.5000 4.0000 4.5000  
5.0000
```

Defining a matrix

```
>> M = [1 2 3; 3 2 1]
```

M =

1 2 3

3 2 1

The FOR loops can be nested

```
>> H = zeros(5);  
for k=1:5  
for l=1:5  
H(k,l) = 1/(k+l-1);  
end  
end  
>> H
```

H =

1.0000	0.5000	0.3333	0.2500	0.2000
0.5000	0.3333	0.2500	0.2000	0.1667
0.3333	0.2500	0.2000	0.1667	0.1429
0.2500	0.2000	0.1667	0.1429	0.1250
0.2000	0.1667	0.1429	0.1250	0.1111

1.4. Creating functions using m-files

Files that contain a computer code are called the *m-files*. There are two kinds of m-files: the *script files* and the *function files*.

- Script files do not take the input arguments or return the output arguments.
- The function files may take input arguments or return output arguments.

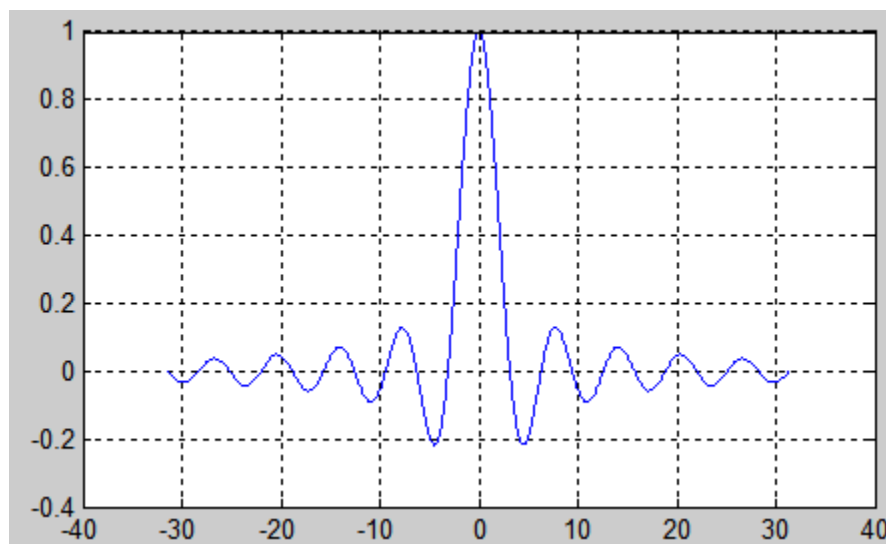
To make the **m-file**;

- Click on **File** next select **New** and click on **M-File** from the pull-down menu. You will be presented with the **MATLAB Editor/Debugger** screen.
- Here you will type your code, can make %changes, etc.
- Once you are done with typing, click on **File**, and select **Save As...**
- Chose a name for your file, e.g., **firstgraph.m** and click on **Save**.

Make sure that your file is saved in the directory that is in MATLAB's search path.

Here is an example of a small script file

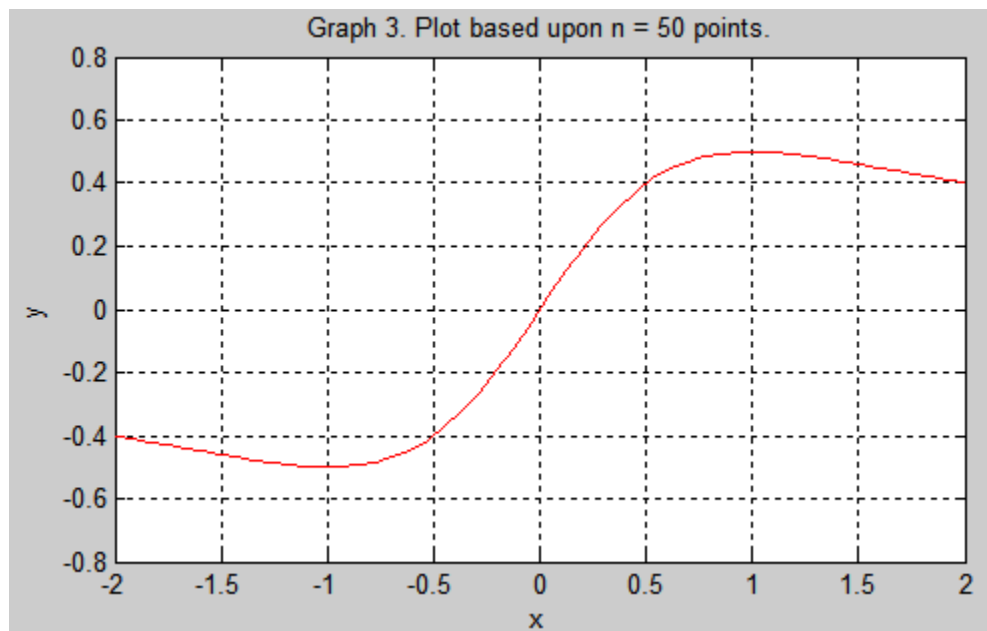
```
x = -10*pi:pi/100:10*pi;  
y = sin(x)./x;  
plot(x,y)  
grid
```



1.5. Graphs in MATLAB

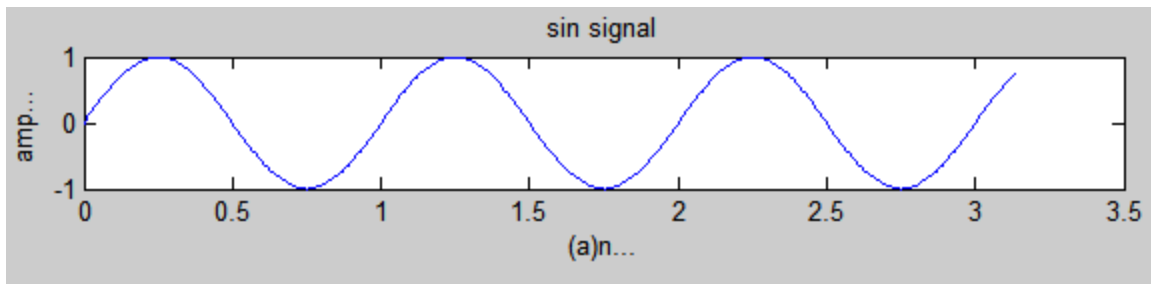
Plot a graph of the rational function $y = x/(1+x^2)$

```
for n=1:2:5
    n10 = 10*n;
    x = linspace(-2,2,n10);
    y = x./(1+x.^2);
    plot(x,y,'r')
    title(sprintf('Graph %g. Plot based upon n = %g points.' ...
        , (n+1)/2, n10))
    axis([-2,2,-.8,.8])
    xlabel('x')
    ylabel('y')
    grid
    pause(3)
end%
```



1.6. Continuous time signal

```
clc;
close all;
clear all;
t=0:.01:pi;
y= sin(2*pi*t);
subplot(4,1,1);
plot(t,y);
ylabel('amp...');xlabel('(a)n...');title('sin signal')
```



1.7. Program for Discrete time cosine signal

```
t=0:.03:pi/3;  
y= cos(2*pi*t);  
subplot(4,1,2);  
stem(t,y);  
xlabel('a(n)');ylabel('amplitude');title('cosinusoidal');
```

